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Structure and fabrication method of high voltage MOSFET with a vertical drift region

Abstract

Embodiments of the present disclosure include a transistor with a vertical drift region and methods for forming the transistor. The transistor may include a well region of a first conductivity type, a gate region disposed above the well region, and a drift region of a second conductivity type, different from the first conductivity type. The drift region may have a lateral portion disposed above a portion of the well region and laterally adjacent to a semiconductor channel in the well region. The drift region may also have a vertical portion extending vertically from the lateral portion of the drift region.

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Background/Summary

BACKGROUND

Field

(1) Embodiments of the present disclosure generally relate to semiconductor fabrication, and more particularly, to systems and methods of forming high voltage metal-oxide-semiconductor field-effect transistors (MOSFETs).

Description of the Related Art

(2) The metal-oxide-semiconductor field-effect transistor (MOSFET) is a type of field-effect transistor (FET). It has an insulated gate, whose voltage determines the conductivity of the device. This ability to change conductivity with the amount of applied voltage is used for amplifying or switching electronic signals.

(3) The MOSFET is by far the most common transistor in digital circuits, as hundreds of thousands or millions of MOSFETs may be included in a memory chip or microprocessor. Since MOSFETs can be made with either p-type or n-type semiconductors, complementary pairs of MOS transistors can be used to make switching circuits with very low power consumption, in the form of complementary metal-oxide-semiconductor (CMOS) logic.

(4) High voltage transistors (e.g., with off-state breakdown voltages (VBD) of 5-30 V) are widely used in a variety of applications, including in NAND flash memory periphery circuits. A NAND flash memory chip may be composed of a cell array and peripheral circuits including X-decoder, Y-decoder, and other circuits. To reduce the size of the cell array for a given memory density, the cell array may have a 3-dimensional structure (e.g., 3D NAND). For each generation of 3D NAND flash memory, the number of memory stacks has been increased. However, the scaling-down of peripheral circuits has been much slower than the increase of memory density. More than half of the area of the peripheral circuits may be occupied by the high voltage transistor region. However, reducing the dimensions of high voltage transistors while maintaining a high VBD has proved challenging.

(5) Thus, there is a need for high voltage transistors that occupy reduced area while maintaining a high VBD, and methods of producing such transistors.

SUMMARY

(6) The present disclosure generally relates to structure and fabrication methods of a high voltage MOSFET with a drift region having a vertical portion.

(7) Embodiments of the present disclosure provide a transistor comprising a well region of a first conductivity type, a gate region disposed above the well region, and a drift region of a second conductivity type, different from the first conductivity type, the drift region comprising a lateral portion disposed above a portion of the well region and laterally adjacent to a semiconductor

channel in the well region, and a vertical portion extending vertically from the lateral portion of the drift region. The transistor may further comprise a drain region or a source region of the second conductivity type. In some embodiments, the vertical portion of the drift region has a first length that extends in a first direction between a surface of the lateral portion and the drain region or extends in a first direction between a surface of the lateral portion and the source region, the lateral portion of the drift region has a second length that extends in a second direction between an edge of the semiconductor channel and an edge of the vertical portion of the drift region, and the sum of the first length and the second length are configured to achieve a breakdown voltage of the transistor of at least 30 V.

(8) Embodiments of the present disclosure provide a method of forming a transistor, the method comprising forming a well region of a first conductivity type, forming a gate region disposed above the well region, and forming a drift region of a second conductivity type, different from the first conductivity type. According to certain embodiments, forming the drift region comprises forming a lateral portion disposed above a portion of the well region and laterally adjacent to a semiconductor channel in the well region, and forming a vertical portion extending vertically from the lateral portion of the drift region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.

(2) FIG. 1A is a cross-sectional view of an example n-type transistor with a drift region having a vertical portion, according to one or more of the embodiments described herein.

(3) FIG. 1B is a cross-sectional view of an example p-type transistor with a drift region having a vertical portion, according to one or more of the embodiments described herein.

(4) FIGS. 2A-2H and 2J-2M are cross-sectional views of intermediate structures at various stages of semiconductor processing to form a transistor structure, according to one or more of the embodiments described herein.

(5) FIG. 3 depicts a top-down view of an example transistor structure, according to one or more of the embodiments described herein.

(6) FIG. 4 depicts a process flow diagram that illustrates a plurality of activities that can be performed to form at least part of a high voltage transistor structure, according to one or more of the embodiments described herein.

(7) FIG. 5 is a simplified schematic example of a 3D NAND structure, in which embodiments of the present disclosure may be implemented.

(8) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(9) In the following description, details are set forth by way of example to facilitate an understanding of the disclosed subject matter. It should be apparent to a person of ordinary skill in the field, however, that the disclosed implementations are exemplary and not exhaustive of all possible implementations. Thus, it should be understood that reference to the described examples is

not intended to limit the scope of the disclosure. Any alterations and further modifications to the described devices, instruments, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one implementation may be combined with the features, components, and/or steps described with respect to other implementations of the present disclosure. As used herein, the terms “about” and “approximately” may refer to a $\pm 10\%$ variation from the nominal value. It is to be understood that such a variation can be included in any value provided herein.

(10) The embodiments described herein provide systems and methods for forming high voltage transistors with drift regions having a vertical portion. The breakdown voltage (VBD) of a transistor is directly proportional to the length of the drift region of the transistor. Accordingly, reducing the dimensions of the transistor by reducing the length of the drift region reduces the VBD. However, by forming part of the drift region vertically, the overall lateral area (e.g., the “footprint”) of the transistor may be reduced without reducing the length of the drift region. Accordingly, the lateral area of the transistor may be reduced without reducing the VBD of the transistor.

(11) FIG. 1A is a cross-sectional view of an example n-type transistor **100A** with drift regions having a vertical portion, according to one or more of the embodiments described herein.

(12) The transistor **100A** may include a p-type well region **102** with an active region defined by isolation regions **104**. The transistor **100A** may also include drift regions defined in the active region. The drift regions may include lightly doped n-type regions (e.g., n⁻ regions), including lateral portions **106** and vertical portions **108**. As shown, the lateral portions **106** of the drift regions may be laterally adjacent to a semiconductor channel **120** in the well region **102**. The lateral portions **106** of the drift regions may have a height (H.sub.1), which represents the depth to which the n-type dopants were implanted in the well region **102**. The height H.sub.1 may have a value between 0.1 μm to 0.3 μm . The vertical portions **108** of the drift regions may have a width (w) and a height (H.sub.2). In some examples, the width w may be approximately equal to the height H.sub.1. In some examples, the height H.sub.2 may have a value between 0.1 μm and 0.6 μm . Although each of the vertical portions **108** as illustrated is disposed in the middle of a lateral portion **106**, a vertical portion **108** may be disposed at a position offset from the middle of a lateral portion **106** or a vertical portion **108** may be disposed at one end of a lateral portion **106**. To avoid confusion relating a definition of a total length of the drift region, the vertical portion, which accounts for a portion of the total length of the drift region, is also referred to as having a “length.” The “length” of the vertical portion **108**, **126** is generally defined as being equal to the height H.sub.2, which extends between the source/drain **114**, **128** and a surface **106A**, **124A** of the lateral region **106**, **124**.

(13) According to certain embodiments, the vertical portions **108** of the drift regions may have a graded doping profile, in which the doping concentration increases as a distance from the well region **102** increases (e.g., as the location in the Z-direction increases). In certain embodiments, the graded doping profile may be graded from a first doping concentration (C1) to a second doping concentration (C2). In certain embodiments, C2 may be 1.1 to 100 times greater than C1. For example, C1 may be about 5×10^{17} atoms per cubic centimeter (atoms/cm³) and C2 may be about 2×10^{19} atoms/cm³. The graded doping profile serves to further increase the breakdown voltage of the transistor **100A**.

(14) The transistor **100A** may include source/drain regions **114** disposed above the vertical portions **108** of the drift regions, and electrically conductive contacts **116** disposed above the source/drain regions **114**. The source/drain regions **114** may be heavily doped n-type regions (e.g., n⁺ regions), as shown. The transistor **100A** may include a gate structure **110**, which may include a gate stack and spacers. The transistor **100A** may also include an inter-layer dielectric (ILD) layer **118**, which acts to provide electrical insulation.

(15) The drift regions may serve as a resistive buffer between the source/drain regions **114** while the transistor **100A** is in an OFF state. That is, the drift regions may prevent (or at least reduce) current flow (e.g., leakage) between the source/drain regions **114** when a voltage below a threshold voltage is applied to the transistor **100A**. When a voltage at or above the threshold voltage is applied to the transistor **100A**, the transistor **100A** is in an ON state, and current may flow through the drift regions between the source/drain regions **114**.

(16) FIG. 1B is a cross-sectional view of an example p-type transistor **100B** with drift regions having vertical portions **126**, according to one or more of the embodiments described herein. The transistor **100B** may be similar to the transistor **100A**, but with certain structures having an opposite conductivity type from those described with respect to FIG. 1A. For example, the transistor **100B** may include a substrate with an n-type well region **122** (as opposed to the p-type well region **102** in transistor **100A**). The transistor **100B** may also include drift regions comprising lightly doped p-type regions (e.g., p⁻ regions), including lateral portions **124** and vertical portions **126**. The transistor **100B** may include source/drain regions **128** disposed above the vertical portions **126** of the drift regions. The source/drain regions **128** may be heavily doped p-type regions (e.g., p⁺ regions), as shown.

(17) Transistors **100A** and **100B** may be considered high voltage transistors, and may have a breakdown voltage (VBD) of at least 30 V. In some embodiments, as shown in FIG. 1A, the vertical portion **108** of the drift region has a first length (e.g., height H.sub.2) that extends in a first direction (i.e., Z-direction) between a surface **106A** of the lateral portion **106** and a source/drain region **114**, and the lateral portion **106** of the drift region has a second length (L.sub.1) that extends in a second direction (e.g., X-direction) between an edge of the semiconductor channel **120** and an edge of the vertical portion **108** of the drift region. Similarly, in some embodiments, as shown in FIG. 1B, the vertical portion **126** of the drift region has a first length (e.g., height H.sub.2) that extends in a first direction (i.e., Z-direction) between a surface **124A** of the lateral portion **124** and a source/drain region **128**, and the lateral portion **124** of the drift region has a second length (L.sub.1) that extends in a second direction (e.g., X-direction) between an edge of the semiconductor channel **120** and an edge of the vertical portion **126** of the drift region. In some embodiments, the sum of the first length and the second length, or drift region length (i.e., DRL=H.sub.2+L.sub.1), and/or the doping levels in the vertical portion **108**, **126** and lateral portion **106**, **124**, respectively, are adjusted to achieve a desired breakdown voltage (VBD), such as a VBD of at least 30 V.

(18) FIGS. 2A-2M are cross-sectional views of intermediate structures at various stages of semiconductor processing to form a transistor (e.g., transistor **100A**) according to one or more of the embodiments described herein. The transistor formed according to the method of FIGS. 2A-2M can form part of a larger device, such as a 3D NAND flash memory chip (discussed further with respect to FIG. 5). While the formation of a n-type transistor (e.g., an NMOS transistor such as the transistor **100A**) is discussed with respect to FIGS. 2A-2M, persons having ordinary skill in the relevant art will appreciate that similar methods may be used to form a p-type transistor (e.g., a PMOS transistor such as the transistor **100B**), in which the doped structures of the transistor have a conductivity type opposite those described below.

(19) Referring to FIG. 2A, a well region **202** is formed by well implantation in a substrate. Alternatively, the well region **202** may be a substrate with a desired doping concentration. The dopants in the well region **202** may be p-type dopants. Accordingly, the well region **202** may be referred to as a p-well. In some examples, the p-well may have a boron (B) doping concentration ranging from approximately 1×10¹¹ to 1×10¹³ atoms/cm³.

(20) Referring to FIG. 2B, an active region **202A** may be defined in a portion of the well region **202**. The active area **202A** can be an area defined between isolation regions **204**, such as shallow trench isolations (STIs). In some examples, the depth (D) of the isolation regions **204** may range from approximately 0.02 μm to 2.0 μm in the Z-direction. In some examples, the width (W) of the

isolation regions **204** may range from 0.05 μm to 5 μm .

(21) Referring to FIG. 2C, gate region layers may be formed on top of the well region **202** and isolation regions **204**. The gate region layers may include a gate oxide layer **210**, a gate layer **212**, and a hard mask layer **220**. In some examples, the gate oxide layer **210** may be formed from silicon dioxide (SiO_2), silicon-oxynitride (SiON), or the like, and may have a thickness of approximately 20 nm to 100 nm in the Z-direction. In some examples, the gate layer **212** may be formed from nitrogen and/or silicon, and may have a thickness of approximately 20 nm to 100 nm in the Z-direction. In some examples, the hard mask layer **220** may be a moderate-temperature oxide (MTO) hard mask, and may have a thickness of approximately 50 nm to 200 nm in the Z-direction.

(22) Referring to FIG. 2D, the gate region layers may be patterned to form the gate structure **230**. For example, the gate structure **230** may be patterned using appropriate lithography and etching techniques (e.g., photolithography), which are known in the art.

(23) Referring to FIG. 2E, an n-type dopant may be implanted into the active region of the well region **202** on either side of the gate structure **230** to form lateral portions **206** of a drift regions. The implanted dopant may be phosphorous, arsenic, or the like. The projected range of the drift regions may be between 0.1 μm and 0.5 μm . The dopant concentration of the lateral portions **206** of the drift regions may range from 1×10^{14} to 1×10^{17} atoms/cm³. Accordingly, the lateral portions **206** of the drift regions may be considered a lightly doped n-type region (e.g., n-region). The lateral portions **206** of the drift regions may also define a semiconductor channel **236** under the gate structure **230**.

(24) Referring to FIG. 2F, gate spacers **222** may be formed on each side of the gate structure **230**. The gate spacers **222** may be formed using standard methods of spacer deposition and etch back, which are known in the art. For example, the gate spacers **222** can be formed by conformally depositing (such as by plasma enhanced chemical vapor deposition (PECVD), atomic layer deposition (ALD), or the like) one or more layers of silicon nitride, silicon oxide, silicon oxynitride, silicon carbon nitride, silicon oxycarbide, or the like, and anisotropically etching (such as by a reactive ion etch (RIE)) the one or more layers. In some examples, the gate spacers **222** may have a thickness of approximately 0.1 μm , as measured in the X-direction at the surface of the drift region.

(25) Referring to FIG. 2G, a first inter-layer dielectric (ILD) layer **218** may be formed above the substrate (e.g., well region **202**) and the gate structure **230**. The first ILD layer **218** may be any suitable dielectric material, such as silicon oxide, borophosphosilicate glass (BPSG), phosphosilicate glass (PSG), or the like. The first ILD layer **218** may be deposited by chemical vapor deposition (CVD), furnace chemical vapor deposition (FCVD), or another appropriate deposition process. After deposition, the first ILD layer **218** may planarized, such as by use of a chemical mechanical planarization (CMP) process. In some examples, the first ILD layer **218** may have a thickness of approximately 0.3 μm to 1.0 μm in the Z-direction.

(26) Referring to FIG. 2H, the first ILD layer **218** may be patterned and etched to form openings **232** through the first ILD layer **218** and stopping at the lateral portions **206** of the drift regions. In some examples, the openings **232** may be formed using a selectively opened mask layer (not shown).

(27) Referring to FIG. 2J, in some examples, silicon or another suitable crystal semiconductor may be selectively grown on the opening of the lateral portions **206** of the drift regions to form the vertical portions **208** of the drift regions. The selectively grown silicon material may be a doped silicon material that is formed during the deposition process. Alternatively, the selectively grown silicon may be un-doped silicon that is then implanted with dopants or doped using gas-phase doping methods. The dopant concentration of the vertical portions **208** of the drift regions may range from 1×10^{14} to 1×10^{20} atoms/cm³.

(28) Alternatively, referring to FIG. 2K, in some examples, silicon or another suitable

semiconductor may be deposited (e.g., epitaxially deposited) such that the silicon fills the openings **232** in the ILD layer **218** and forms a layer **216** over the ILD layer **218**. The deposited silicon may then be etched back to form the vertical portions **208** of the drift regions, as shown. The deposited silicon may be doped silicon material that is formed during the deposition process. Alternatively, the deposited silicon may be un-doped silicon that is later implanted with dopants or doped using gas-phase doping methods.

(29) As shown in both FIGS. **2J** and **2K**, a top of the vertical portions **208** of the drift regions may be above (e.g., at a height greater than) a top of the gate oxide layer **210**. In some examples, the top of the vertical portions **208** of the drift regions may also be above a top of the gate layer **212**, as shown.

(30) Referring to FIG. **2L**, the top of the vertical portions **208** of the drift regions may be doped with n⁺ ions (e.g., phosphorous) by either implantation or plasma gas doping to form source and drain regions **214**. In some cases, the dopant concentration of the source and drain regions **214** may range from 1×10^{19} to 5×10^{20} atoms/cm³. Accordingly, the source and drain regions **214** may be referred to as n⁺ regions. The source and drain regions **214** may have a thickness of 0.01 μm to 0.1 μm in the Z-direction. In some cases, the source and drain regions **214** may be formed by the growth of selectively doped silicon on top of the vertical portions **208** of the drift regions.

(31) Referring to FIG. **2M**, a second ILD layer **228**, and a conductive pad **234** may be formed. The second ILD layer **228** may be a similar material as the first ILD layer **218** and may be deposited in the same or a similar manner. A recess may be formed through the first ILD layer **218** and the second ILD layer **228** using any appropriate photolithography and etching process. The conductive pad **234** may be deposited in the recess, such as by CVD, ALD, or physical vapor deposition (PVD). The conductive pad **234** can include a conformal barrier layer **224**, such as titanium nitride (TiN), tantalum nitride (TaN), or the like, and a metal fill **226** on the barrier layer, such as tungsten (W), aluminum (Al), copper (Cu), or the like. The conductive pad **234** may form an ohmic contact with the respective source/drain region **214** that the conductive pad **234** contacts.

(32) Additional processing steps, such as planarization, metallization, and the like, may be subsequently performed to complete fabrication of the device in which the transistor is formed.

(33) FIG. **3** depicts a top-down view of the transistor **200** of FIG. **2M**, according to one or more of the embodiments described herein.

(34) As shown, an edge of the vertical portions **208** of the drift regions may be spaced a distance **312** away from the semiconductor channel, which is disposed below the gate structure **230** of the transistor (e.g., the semiconductor channel **236** in FIG. **2E**; or from an edge of the lateral portions **206** of the drift regions). Another edge of the vertical portions **208** of the drift regions may be spaced a distance **314** away from an edge **316** of the active region of the transistor **200** (or from another edge of the lateral portions **206** of the drift regions).

(35) Some traditional transistors have source/drain regions disposed within the lateral portions **206** of the drift regions at the location where the vertical portions **208** of the drift regions are shown. Accordingly, the distance between the source/drain regions and the semiconductor channel of such a traditional transistor represents the entire length of each drift region, which may be approximately 0.6 μm .

(36) Contrastingly, in transistor **200**, each of the distances **312** and **314** may be less than 0.6 μm . The height $H_{\text{sub.2}}$ of the vertical portions **208** of the drift regions provides additional spacing between the semiconductor channel and the source/drain regions. Accordingly, the distances **312** and **314** may be reduced by the height of the vertical portions **208** of the drift regions. For example, if the height $H_{\text{sub.2}}$ of the vertical portions **208** of the drift regions is 0.5 μm , the distances **312** and **314** may each be 0.1 μm , as opposed to 0.6 μm . Accordingly, the width of the transistor **200** may be decreased by an amount equal to 4 times the height of the vertical portions **208** of the drift regions. The distances **312** and **314** need not be equal.

(37) FIG. 4 depicts a process flow diagram that illustrates a method **400** of forming at least part of a high voltage transistor structure, according to one or more of the embodiments described herein. Various different examples are described below. Although multiple features of different examples may be described together in a process flow, the multiple features can each be implemented separately or individually and/or in a different process flow. Additionally, various process flows are described as being performed in an order; other examples can implement process flows in different orders and/or with more or fewer operations. Additionally, although source and drain nodes and source and drain regions are described separately in various examples, such description can more generally be to a source/drain node or source/drain region.

(38) The method **400** may begin, at **402**, with forming a well region (e.g., well region **202**) of a first conductivity type. At **404**, the method **400** may involve forming a gate region (e.g., gate structure **230**) disposed above the well region.

(39) At **406**, the method **400** may involve forming a drift region of a second conductivity type, different from the first conductivity type. Forming the drift region may involve, at **408**, forming a lateral portion (e.g., lateral portion **206** of the drift region) disposed above a portion of the well region and laterally adjacent to a semiconductor channel in the well region. Forming the drift region may further involve, at **410**, forming a vertical portion (e.g., vertical portion **208**) extending vertically from the lateral portion of the drift region.

(40) According to some examples, the method **400** may further involve forming a drain or a source region of the second conductivity type disposed above the vertical portion of the drift region. In some examples, the lateral portion of the drift region may extend laterally between the drain or the source region and the semiconductor channel.

(41) According to some examples, forming the gate region may involve forming a gate oxide layer disposed above the well region, and forming a gate layer disposed above the gate oxide layer.

(42) FIG. 5 is a simplified schematic example of a three-dimensional (3D) NAND structure **500**, in which embodiments of the present disclosure may be implemented. For example, the 3D NAND structure **500** may include a memory structure **510** and a transistor array **512** below the memory structure **510**.

(43) In one example, the memory structure **510** may include a bit line **502**, source/drain regions **504**, select gate regions **506**, and a plurality of stacked layer pairs **508**, which include word lines **508A** and an interlayer dielectric layer **508B**. The memory structure **510** may include a staircase (not shown) contact scheme for the word lines **508**, which are found on opposing sides of the plurality of stacked layer pairs **508**.

(44) In certain aspects, the transistor array **512** may include an array of power transistors, such as power complimentary metal-oxide semiconductor (CMOS) transistors. One or more of the transistors in the transistor array **512** may include one or more features of the transistors **100A** and/or **100B** and may be formed according to the method **400** described with respect to FIG. 4. For example, as shown, the drift regions of the transistors in the transistor array **512** may include lateral portions **106**, **124** and vertical portions **108**, **126**. Each of the conductive pads **116** of the transistor array **512** can be coupled to a word line **508A** through a staircase contact (not shown) or to a bit line **502** as necessary to control the storage operations within a memory cell within the 3D NAND device. As shown, the vertical portions **108**, **126** of the drift regions may utilize the vertical space **513** in an inter-layer dielectric (ILD) region of the 3D NAND structure **500**.

(45) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A transistor comprising: a well region of a first conductivity type; a gate region disposed above the well region in a first direction; a drain region or a source region of a second conductivity type, different from the first conductivity type; and a drift region of the second conductivity type, comprising: a lateral portion disposed above a portion of the well region in the first direction, and laterally adjacent to a semiconductor channel in the well region in a second direction perpendicular to the first direction; and a vertical portion extending vertically from the lateral portion of the drift region to the drain region or to the source region in the first direction.
 2. The transistor of claim 1, wherein the vertical portion of the drift region comprises a graded doping profile which increases as a distance from the well region increases.
 3. The transistor of claim 2, wherein the graded doping profile is graded from a first doping concentration (C1) to a second doping concentration (C2), C2 being 1.1 to 100 times greater than C1.
 4. The transistor of claim 1, wherein: the lateral portion of the drift region is disposed between the vertical portion and the semiconductor channel.
 5. The transistor of claim 1, wherein: the gate region comprises: a gate oxide layer disposed above the well region, and a gate layer disposed above the gate oxide layer; and a top of the vertical portion of the drift region is above a top of the gate layer.
 6. The transistor of claim 1, wherein: the gate region comprises: a gate oxide layer disposed above the well region, and a gate layer disposed above the gate oxide layer; and a top of the vertical portion of the drift region is above a top of the gate oxide layer.
 7. The transistor of claim 1, wherein a length of the vertical portion of the drift region is directly proportional to a breakdown voltage of the transistor.
 8. The transistor of claim 1, wherein a length of the vertical portion of the drift region is between 0.1 μm and 0.6 μm .
 9. The transistor of claim 1, wherein: the vertical portion of the drift region has a first length that extends in the first direction between a surface of the lateral portion and the drain region, or extends in the first direction between the surface of the lateral portion and the source region, the lateral portion of the drift region has a second length that extends in the second direction between an edge of the semiconductor channel and an edge of the vertical portion of the drift region, and the sum of the first length and the second length are configured to achieve a breakdown voltage of the transistor of at least 30 V.
 10. A three-dimensional NAND flash memory integrated circuit, comprising a plurality of transistors including the transistor of claim 1.
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